

RPG and LLM



Jack Woehr
Seiden Group



pre-columbian ceramic llama
Walters Art Museum

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Our Agenda

1. Running models
2. Calling from `curl`
3. Calling from SQL
4. Calling from Python
5. IBM OSS DbToo
6. Early easy SQLRPGLE
7. SQLRPGLE with DSPF



Ollama



[Discord](#) [GitHub](#) [Models](#)

🔍 Search models



**Get up and running with large
language models.**

Run [DeepSeek-R1](#), [Qwen 3](#), [Llama 3.3](#), [Qwen 2.5-VL](#),
[Gemma 3](#), and other models, locally.

Download ↓

Available for macOS,
Linux, and Windows

DMI: Dell Inc. Precision Tower 5810/0K240Y

```
$ lscpu -p
# The following is the parsable format, which can be fed to other
# programs. Each different item in every column has an unique ID
# starting usually from zero.
# CPU,Core,Socket,Node,,L1d,L1i,L2,L3
0,0,0,0,,0,0,0,0
1,1,0,0,,1,1,1,0
2,2,0,0,,2,2,2,0
3,3,0,0,,3,3,3,0
4,4,0,0,,4,4,4,0
5,5,0,0,,5,5,5,0
6,6,0,0,,6,6,6,0
7,7,0,0,,8,8,8,0
8,8,0,0,,9,9,9,0
9,9,0,0,,10,10,10,0
10,10,0,0,,11,11,11,0
11,11,0,0,,12,12,12,0
12,12,0,0,,13,13,13,0
13,13,0,0,,14,14,14,0
14,0,0,0,,0,0,0,0
15,1,0,0,,1,1,1,0
16,2,0,0,,2,2,2,0
17,3,0,0,,3,3,3,0
18,4,0,0,,4,4,4,0
19,5,0,0,,5,5,5,0
20,6,0,0,,6,6,6,0
21,7,0,0,,8,8,8,0
22,8,0,0,,9,9,9,0
23,9,0,0,,10,10,10,0
24,10,0,0,,11,11,11,0
25,11,0,0,,12,12,12,0
26,12,0,0,,13,13,13,0
27,13,0,0,,14,14,14,0
```

```
$ nvidia-smi
Sat Jul 19 10:54:17 2025
+-----+
| NVIDIA-SMI 535.247.01              Driver Version: 535.247.01   CUDA Version: 12.2   |
+-----+-----+-----+-----+-----+-----+
| GPU  Name            Persistence-M   Bus-Id        Disp.A    Volatile Uncorr. ECC |
| Fan  Temp            Perf             Pwr:Usage/Cap     Memory-Usage | GPU-Util  Compute M. |
|                                           MIG M.         |
+-----+-----+-----+-----+-----+-----+
|   0   NVIDIA GeForce GTX 1650      Off          00000000:02:00:00 Off      N/A
| 27%   29C            P8              5W / 75W          49MiB / 4096MiB      0%      Default
|                                           N/A               |
+-----+-----+-----+-----+-----+-----+

+-----+
| Processes:
| GPU   GI    CI          PID    Type    Process name                        GPU Memory
|      ID    ID                                   Usage
+-----+-----+-----+-----+-----+-----+
|   0   N/A    N/A         2132     G      /usr/lib/xorg/Xorg                  39MiB
|   0   N/A    N/A         2318     G      /usr/bin/gnome-shell                5MiB
+-----+

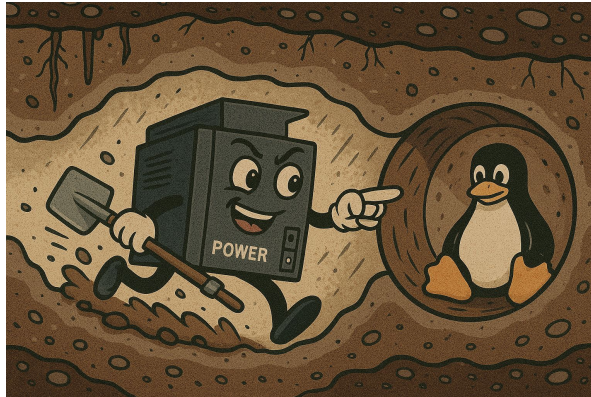
$ lsmem
RANGE                                SIZE  STATE REMOVABLE BLOCK
0x0000000000000000-0x000000007fffffff  2G online      yes    0
0x0000000100000000-0x00000014ffffffff  80G online      yes  2-41

Memory block size:      2G
Total online memory:    82G
Total offline memory:   0B
```

Reverse tunnel to Linux from IBM i

- Allow ourselves to use Ollama running on the Linux server from IBM i making the port appear local.
 - Necessary because the Linux server does not have a publicly accessible IP address.
- Leave this up as long as we are going to use Ollama from IBM i.

```
ssh -R 11434:localhost:11434 -l myuserprof mysystem.mycompany.com
```



curl the query

```
curl http://localhost:11434/api/chat -d '{
  "model": "llama3.2:3b",
  "stream": false,
  "messages": [
    { "role": "user", "content": "why do dogs bark?" }
  ]
}'
```



Patrick Behr calls ChatGPT from SQL using HTTP_POST

```
CREATE OR REPLACE FUNCTION pbehr.askChatGpt
```

```
(  
    in_prompt      varchar(1024)  
    , in_sysmsg    varchar(512)    default null  
    , in_temp      varchar(10)     default '1'  
    , in_maxtokens  varchar(10)     default '500'  
)
```

```
RETURNS VARCHAR(4096)
```

```
BEGIN
```

```
    DECLARE apiUrl  VARCHAR(64)  DEFAULT  
'https://api.openai.com/v1/chat/completions';  
    DECLARE apiKey  VARCHAR(64)  DEFAULT  
'sk-pd38i1R9XAMDVvk34Key4T3BlbkFJuzIV5kX8tKUainKAs4XG';  
    DECLARE model   VARCHAR(32)  DEFAULT 'gpt-3.5-turbo';  
    DECLARE headers VARCHAR(256)  DEFAULT '';  
    DECLARE messages VARCHAR(2500) DEFAULT '';  
    DECLARE sysmsg  VARCHAR(1000) DEFAULT '';
```

```
    DECLARE userMsg  VARCHAR(1500) DEFAULT '';  
    DECLARE request  VARCHAR(4096) DEFAULT '';
```

```
    SET headers = '{"headers":{"Authorization":  
"Bearer ' CONCAT apiKey CONCAT '",'
```

```
"Content-Type":"application/json"}}';  
  
    IF in_sysmsg IS NOT NULL THEN  
        SET sysmsg = '{"role":"system",  
"content":"' CONCAT in_sysmsg CONCAT '"},"';  
    END IF;
```

```
    SET userMsg = '{"role":"user", "content":'  
CONCAT in_prompt CONCAT '"}';
```

```
    SET messages = '[' CONCAT sysmsg CONCAT  
userMsg CONCAT ']';
```

```
    SET request = '{"model":"'          CONCAT  
model          CONCAT '",'          CONCAT  
                "temperature":"'      CONCAT  
in_temp        CONCAT '",'          CONCAT  
                "max_tokens":"'        CONCAT  
in_maxtokens   CONCAT '",'          CONCAT  
                "messages":"'          CONCAT  
messages       CONCAT '"}';
```

```
    RETURN (  
        SELECT apiResponse  
        FROM JSON_TABLE(  
            HTTP_POST( apiUrl, request, headers  
        ),  
            'lax $.choices[0]'  
            COLUMNS(  
                apiResponse varchar(4096) PATH  
'lax $.message.content'  
            )  
        )  
    );  
  
END;
```

Same sort of thing in Python

```
import ollama
# response = ollama.chat(model='llama3.1:8b', messages=[
# response = ollama.chat(model='granite-code:8b', messages=[
response = ollama.chat(model='llama3.2:3b', messages=[
    {
        'role': 'user',
        'content': 'Why is the sky blue?',
    },
])
print(response['message']['content'])
```



DbToo

- Code: <https://github.com/IBM/AI-SDK-Db2-IBMi/>
- Docs: <https://ibm.github.io/AI-SDK-Db2-IBMi/>

The screenshot shows the documentation page for DbToo, titled "A Db2 for i SDK for connecting to various services". The page has a search bar at the top right with the text "Search" and a "Ctrl K" shortcut. On the left, there is a navigation menu with the following items: Home (highlighted), IBM watsonx, Ollama, OpenAI-compliant endpoints, Kafka, PASE, Slack, SMS (Twilio), and Other Useful links (external) which includes IBM. The main content area is titled "DbToo" and "Overview". It contains a welcome message: "Welcome! This projects aims to simplify the task of integrating RPG and/or Db2 workloads with external services. The navigation pane on the left lists technologies supported by this experimental SDK." Below this is a "Note" box stating: "This project is still in bringup state and should not be deployed to production!". The "Installation" section follows, with a "Caution" box stating: "The installing user profile MUST have the QIBM_DB_SECADM function usage authority!". Below the caution box, it says: "To install on IBM i, you need to clone this git repository onto an IBM i. You can do this with the following methods:" and lists two steps: 1. use `git clone` on this repo, 2. download the repo as a .zip from GitHub and extract it to your IBM i. At the bottom, it says: "Once the repo sources are in the IFS, then you can use GNU Make to build the SDK. GNU Make,".

A Db2 for i SDK for connecting to various services

Search Ctrl K

Home

IBM watsonx >

Ollama >

OpenAI-compliant endpoints >

Kafka >

PASE >

Slack >

SMS (Twilio) >

Other Useful links (external) v

IBM

DbToo

Overview

Welcome! This projects aims to simplify the task of integrating RPG and/or Db2 workloads with external services. The navigation pane on the left lists technologies supported by this experimental SDK.

Note

This project is still in bringup state and should not be deployed to production!

Installation

Caution

The installing user profile MUST have the QIBM_DB_SECADM function usage authority!

To install on IBM i, you need to clone this git repository onto an IBM i. You can do this with the following methods:

1. use `git clone` on this repo
2. download the repo as a .zip from GitHub and extract it to your IBM i

Once the repo sources are in the IFS, then you can use GNU Make to build the SDK. GNU Make,

On this page

[Overview](#)

[Overview](#)

[Installation](#)

Do it in DbToo

```
set
  dbsdk_v1.ollama_protocol = 'http';
set
  dbsdk_v1.ollama_server = 'localhost';
set
  dbsdk_v1.ollama_port = 11434;
set
  dbsdk_v1.ollama_model = 'llama3.2:3b';
select
  dbsdk_v1.ollama_getmodel ()
from
  SYSIBM.SYSDUMMY1;
```

```
select
  dbsdk_v1.ollama_generate (
    prompt => '{ "role": "user", "content": "why do dogs
bark?"}') as myanswer
from
  SYSIBM.SYSDUMMY1;
set
  dbsdk_v1.ollama_model = 'granite-code:8b';
select
  dbsdk_v1.ollama_generate (
    prompt => '{ "role": "user", "content": "why do dogs
bark?"}') as myanswer
from
  SYSIBM.SYSDUMMY1;
```

Do it in SQLRPGLE (v-e-r-y simply)

```
**FREE
// Jack Woehr jack@seidengroup.com 2025-06-09
Ctl-Opt DftActGrp(*No) Option(*Srcstmt : *NodebugIO) main(qOllama);

dcl-s MyAnswer SQLTYPE(clob:16000000);
dcl-s DspStr varchar(52);

dcl-proc qOllama;

    exec sql call dbsdk_v1.ollama_setmodelforme (model => 'granite-code:8b');
    exec sql select dbsdk_v1.ollama_generate (prompt => '{ "role": "user", "content": "why do dogs bark?" }' ) into :MyAnswer
from SYSIBM.SYSDUMMY1;
    DspStr = MyAnswer_DATA;
    dsply DspStr;

    *inlr=*on;
end-Proc;
```

Roo, based on my example

```
**FREE
// Jack Woehr jack@seidengroup.com 2025-09-26
// New RPG program to interact with an Ollama-compatible LLM API
// QSYS/CRTSQLRPGI OBJ(NEWLLM) SRCSTMF('/home/jwoehr/work/AI/COMMONSweden25Q4/code/rpgollama/NEWLLM.SQLRPGLE') TEXT('Roo-authorized take on RPGOLLAMA') CVTCCSID(*JOB)
Ctl-Opt DftActGrp(*No) Option(*Srcstmt : *NodebugIO) main(qNewLLM);

dcl-s MyAnswer SQLTYPE(clob:16000000);
dcl-s DspStr varchar(52);
dcl-s PromptText varchar(200) inz('{ "role": "user", "content": "Tell me a short story about a brave knight."}');

dcl-proc qNewLLM;
// Configure Ollama server settings
exec sql call dbsdk_v1.ollama_setserverforjob(hostname => 'localhost');
exec sql call dbsdk_v1.ollama_setportforjob(port => '11434');
exec sql call dbsdk_v1.ollama_setprotocolforjob(protocol => 'http');
exec sql call dbsdk_v1.ollama_setmodelforjob(model => 'llama3.2:3b'); // Using llama3.2:3b as per user's available model

// Generate response from LLM
exec sql select dbsdk_v1.ollama_generate(prompt => :PromptText) into :MyAnswer from SYSIBM.SYSDUMMY1;
if SQLSTATE = '00000';
    DspStr = MyAnswer_DATA;
    dsply DspStr;
    snd-msg *INFO DspStr %target(*CALLER);
else;
    DspStr = 'SQL Error: ' + SQLSTATE;
    dsply DspStr;
    snd-msg *INFO DspStr %target(*CALLER);
endif;

*inlr=*on;
end-Proc;
```

AI via DSPF

- OLLAMARPG
- GEMQRY

Enter the desired values directly or press F4 to prompt.
If the value doesn't fit below, add the value to the config and use the prompt.

Protocol: http

Server: localhost

Port: 11434

Model: gpt-oss:20b

F12=Cancel F6=Make My Default Enter=Use these values

MA + B 06/014

AI via DSPF

- **OLLAMARPG**
- GEMQRY

```
URL: http://localhost:11434
Model: gpt-oss:20b

Prompt: What is the name of the famous elephant of french childrens' fiction?

Resp: The famous elephant is **Babar** ? the beloved character created by
Jean de Brunhoff in the French children's books of the 1930s.

F3=Exit  F4=Config

MA + B
```

06/010

AI via DSPF

- OLLAMARPG
- **GEMQRY**

```
// Call the GEMINI_QUERY function
EXEC SQL

SET :hResult = GEMINI_QUERY(
    MODEL => TRIM(:hModel),
    KEY => TRIM(:hKey),
    CONTENT => TRIM(:hContent)
);
```

Call Gemini Query w/DTAARA (CALLGEMQYD)

Type choices, press Enter.

Gemini model name 'gemini-3.1-pro-preview'

Query text 'What is IBM i?'

Bottom

F3=Exit F4=Prompt F5=Refresh F12=Cancel F13=How to use this display
F24=More keys

MA + B 05/061

AI via DSPF

- OLLAMARPG
- **GEMQRY**

```
// Call the GEMINI_QUERY function
EXEC SQL
SET :hResult = GEMINI_QUERY(
    MODEL => TRIM(:hModel),
    KEY => TRIM(:hKey),
    CONTENT => TRIM(:hContent)
);
```

```
Display All Messages

Job . . . : JWOEHRB0      User . . . : JWOEHR      System: SEIDEN
Number . . . : 819618

? *N
{? "candidates": [{?      {?      "content": {?      "parts": [{?
  {?      "text": "**IBM i** is an operating system (OS) developed
by IBM, designed specifically for enterprise-level business computing.
It currently runs on **IBM Power Systems** servers alongside other
operating systems like Linux and AIX. \n\nTo understand IBM i, it helps
to know its defining characteristics, its history, and why thousands of
businesses around the world still rely on it today.\n\nHere is a
breakdów
? *N
  Use DSPJOBLOG to see full result
? *N
3> wrkmsg
3>> dspjoblog

Press Enter to continue.

F3=Exit  F5=Refresh  F12=Cancel  F17=Top  F18=Bottom

MA + B 05/001
```


The code (and the PDFs for the presentation)



Thank you



Any sufficiently advanced technology is indistinguishable from magic. - A. C. Clarke



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